

Dolphin

Camera-based autonomous driving.

From driving data to on-vehicle experiments. We develop end-to-end driving systems and explore reasoning, simulation and efficient learning for autonomous driving.

CURRENT STATUS

E2E: on-vehicle proof of concept.

VLA: early integration; shadow-mode evaluation planned.



Phenikaa-X research vehicle

From data to vehicle evaluation



Dolphin Surround-View Explorer: camera views, annotations and BEV.

01 Data collection & curation

Capture, calibration, timestamp alignment, semi-automatic annotation and visualization.

02 Driving model development

Perception, prediction and planning, progressing toward E2E driving and VLA guidance.

03 Real-time integration

PhenikaaX Drive Platform, efficient communication and on-vehicle execution.

04 Simulation & evaluation

HD-map-based and camera-based simulation development, closed-loop evaluation and controlled tests.

Scientific questions with driving applications

01 Efficient end-to-end driving

Architectures and temporal representations under practical compute, latency and memory constraints.

02 Multimodal reasoning & VLA

Spatial and temporal reasoning, useful driving guidance, and decision-relevant datasets and benchmarks.

03 Data-efficient perception & learning

Data selection, limited-label learning and adaptation across traffic scenarios and operating conditions.

04 4D reconstruction & simulation

Improving geometric detail, visual quality and temporal consistency in dynamic scenes for simulation and closed-loop evaluation.

05 Learning better driving decisions

Trajectory diversity and policy improvement from data, with reduced dependence on dedicated simulators.

Selected ongoing and planned directions; related research topics remain open for discussion.

Selected project videos



End-to-end driving

On-vehicle proof of concept.

Watch video



Data visualization

Driving data and annotation workflow.

Watch video



Ocean Park AV

An autonomous vehicle demonstration.

Watch video

IN THE MEDIA

Dolphin on VTV

Dolphin on Hanoi TV

Select a title or button to watch.

Contact us

We welcome research collaborations, advisory partnerships and student research opportunities in autonomous driving. Our team contributes implementation, experiments and vehicle integration.

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CORE TEAMS

E2E / Perception · Reasoning / Planning · Sensing / System
Navigation · Localization · Control

CONTACT

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